Pranjal Kalekar

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SUMMARY

Applied AI engineer with strong expertise in designing and deploying scalable machine learning models using Python and contemporary toolkits. Developed graph-based imputation algorithms and optimized GNN and XGBoost models to achieve up to 40% performance gains over baselines in complex datasets. Collaborative problem solver adept at driving data-driven solutions from experimentation to production.

TECHNICAL SKILLS

- **Tech Languages:** Python, MySQL, MS SQL, R, MATLAB, No-SQL, C++, HTML/CSS
- **Databases:** MongoDB, MariaDB, Neo4j, Apache Spark, PostgreSQL, NoSQL, RDBMS
- **Tools:** Git, VS code, PyCharm, Google Cloud Platform, Tableau, PowerBI, WordPress, Azure, Hadoop
- **Libraries:** PyTorch, Plotly, Tensorflow, SciPy, pandas, Scikit-learn, dlib, seaborn, NLTK, XGBoost, NetworkX
- **Platforms:** Microsoft Excel, Google Colab, PySpark, MS Office, Unix/Linux, MySQLWorkbench, R Studio
- **AI & Machine Learning:** AI/ML Experience, Machine Learning Techniques

EDUCATION

Northeastern University | ***Master of Science, Data Analytics Engineering*** **Sep 2022 - Dec 2024**

VJTI | ***B.Tech, Electronics Engineering*** **Aug 2018 - May 2022**

EXPERIENCE

Haas Lab, Mass General Hospital- Harvard Medical School

Research Engineer *Boston, MA* · **Mar 2025 - Present**

- Engineered a graph-based imputation, reducing data sparsity and achieving a 0.07 RMSE (35% better than baselines)
- Deployed GPU-optimized XGBoost and Neural Networks, enhancing accuracy by 40% over linear models and isolating critical biological feature importance.
- Stabilized GNN training through architectural refactoring and hyperparameter tuning, exceeding performance benchmarks by 25%.

Data Analyst-1 *Boston, MA* · **Jan 2024 - Sep 2024**

- Applied **PCA, UMAP, TSNE, and DBSCAN** to mass-spec proteomics data, improving data quality and identifying distinct breast cancer subtypes.
- **Built GNNs on protein-association networks** (302 databases) using NetworkX, addressing class imbalance to maximize model performance. **Delivered 96% AUC-ROC** on node classification models and presented findings to multi-disciplinary research groups.

DataZ

Associate *Remote – Boston, MA* · **Jan 2025 - Mar 2025**

- Built Python-based web scraping pipelines using Selenium, Scrapy, BeautifulSoup and ScraperAPI for specific product websites similar to Alibaba/AliExpress and centralized 30K+ product entries into a MySQL database, enabling structured analytics and automation.
- Delivered actionable insights through Tableau dashboards tracking key engagement metrics, supporting investor updates and product decisions in a lean startup environment.

Saint Louis University

Data Visualization Intern *Remote* · **Oct 2021 - Nov 2021**

- As a Technical team lead, performed profit focused Meta ads analysis on user performance data, resulting in a monthly subscriber increase of 5% on the client's Meta page. Developed NLP based features by sentiment analysis, sentiment trends based on user interactions via comments.
- Built interactive data visualizations utilizing Tableau, Seaborn, MAtplotlib to support strategic business decisions.

Eddy Tools and Tech Solutions

Computer Vision Intern *Remote* · **Jun 2021 - Sep 2021**

- **Led an 11-member cross-functional team** to deliver a "Live Attention Tracking" POC, meeting all client requirements. **Built a real-time attention monitoring tool** using facial landmarks using Dlib face recognition pipeline and variance analysis.
- **Leveraged Tableau and Power BI** to transform complex datasets into actionable business intelligence and statistical visualizations.

AIRPIX

AI, ML AND EDGE COMPUTING INTERN *Mumbai* · **Jun 2020 - Sep 2020**

- Created a large and well-labeled dataset using Python and YOLO object detection algorithm; effective on low-quality CCTV images.
- Colaborated in training a TensorFlow model to detect PPE in possible threat areas, such as industrial sites and mine areas

PROJECTS

KLCE & GIB Regularization - Class Imbalance Resolution **Sep 2023 - Dec 2023**

- **Integrated KLCE regularization (ICLR 2024)** into GNNs, achieving a 0.9%–29% performance gain over standard log loss functions.
- **Applied Graph Information Bottleneck (GIB)** techniques to the PubMed dataset, increasing balanced accuracy by 10% through optimized relevance filtering. **Architected GNNs using PyTorch Geometric**, implementing complex loss functions and conducting rigorous hyperparameter optimization with Optuna.

CredRisk **Sep 2023 - Dec 2023**

- **Developed a Credit Risk forecasting model** achieving 89% balanced accuracy and a 0.97 specificity score for high-risk loan detection.
- Engineered an ETL pipeline in Python to clean and process LendingClub data, optimizing feature selection via target variable correlation. **Synthesized risk insights** through interactive Tableau dashboards and Python-based visualization libraries.
- **Deployed XGBoost, Random Forest, and Logistic Regression** models, utilizing SMOTE oversampling to mitigate class imbalance.

All-in-one Airport Management System **May 2022**

- Using ETL pipelining Created an RDBMS substantially containing the entities in Airport Management system including terminals, pilots, employees, stores, etc. from different data sources. created/ connected the entities with specific access control
- Got insights into the system using CTEs, window functions, and different MySQL Queries.
- Visualized the database using Tableau dashboard and Python Libraries Such as Matplotlib, Plotly, and Seaborn

CERTIFICATES AND LEADERSHIP

- Hacker at Sundai MIT- Harvard AI Prototype Launching Club, Kaggle Advanced SQL certificate, Former VP of Relations and Projects, Data Science Hub at Northeastern University, Graduate Certification in Leadership affiliated with Northeastern University.
- Resident Assistant at Northeastern University, Climate Justice Fellow 2023, NU cycling- Road Biking ECCC Racing, WRBB Radio Show Host women in mythology, Fiber artist - Pranpaan, MIT- ESP splash Teacher, Mentor at Wellmet transitional housing.